CareerLens — Salary Prediction Engine

COMPREHENSIVE EMPIRICAL ABLATION STUDY REPORT

Platform: CareerLens (career-intelligence-platform) | **Model:** RandomForestRegressor + Quantile Calibrator
Dataset: 7,680 stratified multi-profession profiles | **Split:** 80% Train (6,144) / 20% Held-Out Test (1,536) | **Status:** Evaluated on Test Split

---

1. Executive Summary

To rigorously quantify the individual contribution of each algorithmic module, feature vector dimension, and architectural mechanism in the CareerLens compensation engine, we conducted a systematic two-phase ablation study on a strictly held-out test split of **1,536 unseen profiles**. Our experiments confirm that compensation estimation is fundamentally non-linear across global labor markets, causing linear models to collapse (MAPE > 61%), while domain isolation and geographic market conditioning form the indispensable bedrock of realistic salary predictions.

2. Model Architecture & Capacity Ablation

We compared the full production Random Forest (100 estimators, max depth 14) against constrained ensemble sizes, a single unpruned decision tree, shallow trees, and classic linear baselines (L2 Ridge and OLS).

Architecture / Model	Parameters	Test R^2	MAPE (%)	Test MAE ($)	Test RMSE ($)	Outcome
Full Random Forest	N=100, D=14	**0.9637**	**8.79%**	**$8,517.79**	**$13,810.04**	**Production Baseline**
Random Forest (10 Trees)	N=10, D=14	0.9621	9.10%	$8,710.45	$14,105.78	Slight variance increase
Single Decision Tree	N=1, D=14	0.9385	11.59%	$10,974.25	$17,961.79	+28.8% MAE error surge
Shallow Random Forest	N=100, D=5	0.8918	17.50%	$15,882.48	$23,835.95	Severe underfitting
Ridge Regression (L2)	alpha=1.0	0.3171	61.25%	$40,535.01	$59,873.39	Catastrophic linear failure
Ordinary Least Squares (OLS)	Linear	0.3166	61.22%	$40,545.37	$59,896.54	Incapable of non-linear fit

3. Feature Dimension Ablation (Leave-One-Out Importance)

Each input feature was systematically excluded from the model matrix, retraining under identical random seeds to directly measure accuracy degradation (ΔR^2 and MAE expansion).

Ablated Dimension Dropped	Retrained R^2	MAPE (%)	Retrained MAE ($)	Retrained RMSE ($)	Accuracy Loss	Criticality Level
None (Full 7 Dimensions)	**0.9637**	**8.79%**	**$8,517.79**	**$13,810.04**	**0.00%**	**Optimal Fit**
Without Education	0.9625	9.11%	$8,785.63	$14,027.54	-0.12%	Low / Fine-tuning
Without Skills Premium	0.9596	9.25%	$9,040.18	$14,568.65	-0.41%	Moderate (Niche skills)
Without Experience (Years)	0.9532	9.82%	$9,672.09	$15,669.08	-1.05%	Significant intra-band
Without City Tier	0.9464	10.96%	$10,746.23	$16,777.90	-1.73%	High (Metro vs Regional)
Without Seniority Level	0.9320	11.46%	$11,720.48	$18,900.11	-3.17%	Very High (Band bounds)
Without Domain Isolation	**0.7066**	**30.90%**	**$26,731.43**	**$39,245.54**	**-25.71%**	**Critical (Trade Isolation)**

Without Country	**0.4353**	**95.79%**	**$39,725.08**	**$54,444.57**	**-52.84%**	**Foundational (Macro economy)**
Experience Only (Naive)	0.1595	111.46%	$46,857.70	$66,425.99	-80.42%	Completely inadequate

Key Architectural Insights:

• ***The Critical Role of Domain Isolation:*** *Removing Domain causes MAE to surge from $8.5K to $26.7K (+214%), demonstrating that Universal Role Taxonomy domain mapping is essential to prevent trade professions (Electrician, Plumber) from receiving inflated tech compensation.*

• ***Seniority vs. Experience:*** *Seniority level exerts 3x greater predictive leverage (-3.17% R^2) than chronological experience years (-1.05% R^2), as organizational tier dictates compensation brackets while years dictate positioning within the bracket.*

• ***Non-Linear Dynamics:*** *Linear models (Ridge/OLS) fail catastrophically ($R^2 \approx 0.317$, MAPE > 61%) proving the necessity of tree-based ensemble estimators.*